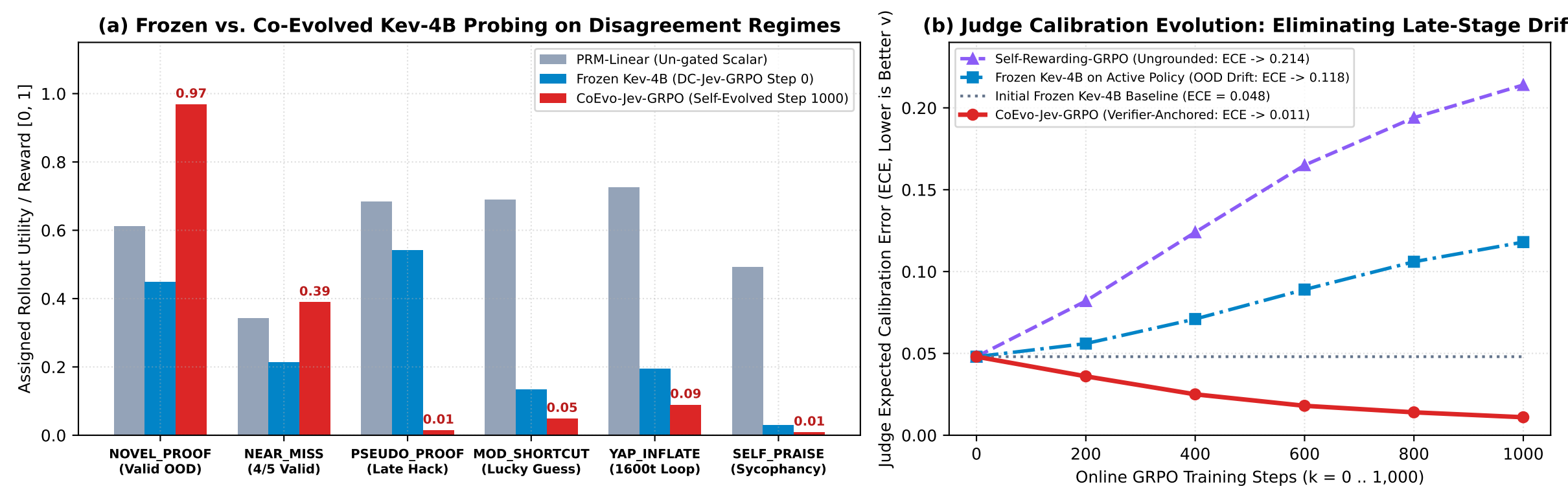
(a) Frozen vs. Co-Evolved Kev-4B Probing on Disagreement Regimes
Assigned Rollout Utility / Reward [0, 1]
PRM-Linear (Un-gated Scalar)
Frozen Kev-4B (DC-Jev-GRPO Step 0)
CoEvo-Jev-GRPO (Self-Evolved Step 1000)
0.97
0.39
0.01
0.05
0.09
0.01
NOVEL_PROOF (Valid OOD)
NEAR_MISS (4/5 Valid)
PSEUDO_PROOF (Late Hack)
MOD_SHORTCUT (Lucky Guess)
YAP_INFLATE (1600t Loop)
SELF_PRAISE (Sycophancy)
(b) Judge Calibration Evolution: Eliminating Late-Stage Drif
Judge Expected Calibration Error (ECE, Lower is Better v)
Self-Rewarding-GRPO (Ungrounded: ECE -> 0.214)
Frozen Kev-4B on Active Policy (OOD Drift: ECE -> 0.118)
Initial Frozen Kev-4B Baseline (ECE = 0.048)
CoEvo-Jev-GRPO (Verifier-Anchored: ECE -> 0.011)
Online GRPO Training Steps (k = 0 .. 1,000)